

Reversing the Mechanism

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We have to take care of these two fields with an unpleasant expectation of connectedness among them. So, why not get a look at something meticulous and start handling with a few pieces of wood that the outer bark.

Nanophotonics and Thermal Electrodynamics Radiations

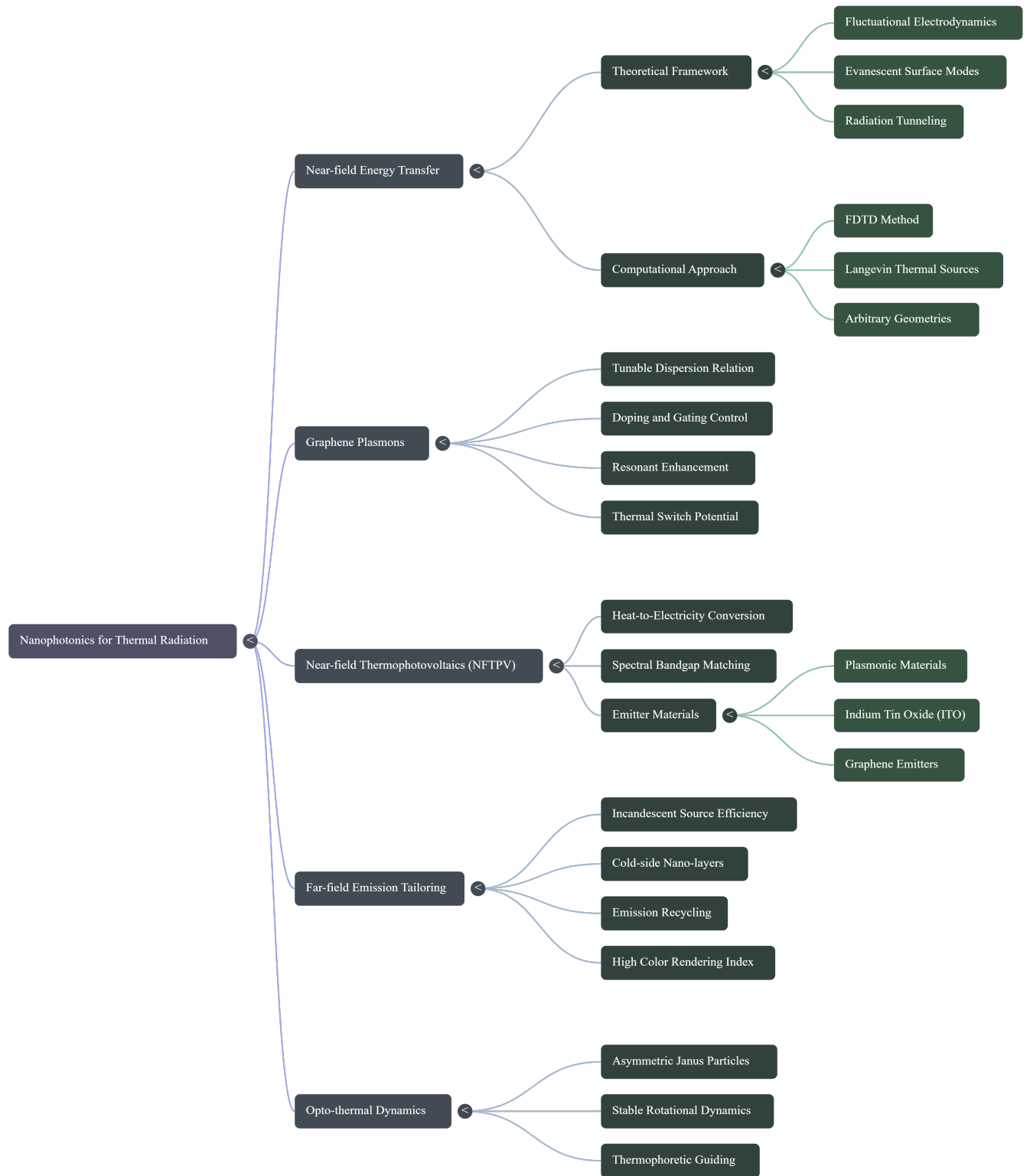
It was a thesis written by **Ognjen Ilic**, on MIT Libraries, about **Nanophotonics for Tailoring the Flow of Thermal Electromagnetic Radiation**.

Our first site of encounter is this thing called **"Nanophotonics"**, which, after some searching, comes out to be the study of light on its behaviour at nanoscale with extremely small objects, it bypasses extreme optical analysis. Which may become useful for us in future. But, in order to breakdown this whole text to a logical basis, we need to construct a mind map. Honestly, I have used Notebooklm preferentially for this work, please don't mind if I do so, as it was not possible for me to handle this text alone. It constructs the following mind-map for our logical framework. It may look awkwardly small, but persist this diagram, as this could be the best thing to deconstruct out text to smaller chunks and get a clearer view of this. A detailed analysis will be done at the end of this study, yet to be soon. I have got a massive plausibly and fundamentally impressive things out of this text. I would like to state them quickly amongst you:

1) The author used **fluctuational electrodynamics**, to get a clear association of our interests. Due to certain temperature of an object, it contains a large amount of charges, these charges move due to thermal energy in a chaotic manner. And according to electrodynamics, movement of charge causes electric current (remember **Faraday**) and so electromagnetic field. This gives us a clear rule:

$$\boxed{\text{Thermal Energy} \rightarrow \text{Chaotic Particle motion} \rightarrow \text{Electromagnetic Field}}$$

This was a good start for our initial idea of interest. **We learned that, to compare those two field, we must deal with charge (field and current), temperature and their association is energy**



2) The exchange of heat or energy took place using thermodynamics, if the objects are relatively far apart, the author explicitly states it to be **far-fields**. But as the distance between objects is decreased

significantly, to a nano-scale, then a type of **evanescent waves** are emerged. These are a type of waves that clings between how objects, and don't evolve outward in space. We use maxwell's equation in this situation.

Far fields $\rightarrow$ Thermodynamics
Evanescent waves $\rightarrow$ Electrodynamics

The heat tranfer at this scale is hundred and thousands of times greater than in the first case. With upcoming study and gained knowledge we will focus on three of the real-world applications that the author dealt with:

-Thermophotovoltaics
-Resurrecting the llight bulb
-Microscopic Motors
